Primitive roots, Zsigmondy and constructions

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L5-NT2 Level 5 Advanced · Number theory

Why primitive roots exist and how to compute with them, Zsigmondy's theorem as a tool, Chinese remainder constructions and proofs that infinitely many primes have a property.
By the end you can prove that primitive roots exist modulo p and p^k, and use indices to count solutions of $x^k \equiv a$; apply Zsigmondy's theorem correctly, or replace the theorem by orders and lifting the exponent; build Chinese remainder constructions and prove that infinitely many primes have a given property.
Prerequisites: Orders, Fermat and Euler (L4-NT1), Olympiad Diophantine equations (L4-NT2), Quadratic residues (L5-NT1). **Contests:** BMO2, EGMO, USAMO, IMO. **Time:** about 180 minutes.

1 Lesson

Compute $S_k = 1^k + 2^k + \cdots + (p-1)^k$ modulo a prime p. For $k = 1$ the sum is $\frac{p(p-1)}{2} \equiv 0$, and for $k = 2$ the sum is $\frac{(p-1)p(2p-1)}{6} \equiv 0$ when $p > 3$, but the closed forms get worse as k grows. Suppose instead that some residue g has the property that $g^0, g^1, \ldots, g^{p-2}$ are all the nonzero residues. Then

$$S_k \equiv \sum_{j=0}^{p-2} g^{jk} \pmod{p}.$$

If $p-1 \mid k$, every term is 1 and $S_k \equiv p-1 \equiv -1$. Otherwise $g^k \not\equiv 1$, and the geometric series gives $(g^k - 1)S_k \equiv g^{k(p-1)} - 1 \equiv 0$, so $S_k \equiv 0$. This answers the question for every k, provided g exists.

Theorem 1 (power sums).
$1^k + 2^k + \cdots + (p-1)^k$ is congruent to -1 modulo p if $p-1 \mid k$, and to 0 otherwise.

Such a g is called a primitive root, and such a root always exists (Theorem 4). The other two tools in this unit are Zsigmondy's theorem on new prime factors of $a^n - b^n$, and constructions with the Chinese remainder theorem. Orders, Euler's theorem, Lagrange's theorem and LTE are from L4-NT1, and Legendre symbols and reciprocity are from L5-NT1.

1.1 Existence of primitive roots

An integer g coprime to n is a **primitive root** modulo n if $\operatorname{ord}_n(g) = \varphi(n)$. Then $g^0, g^1, \ldots, g^{\varphi(n)-1}$ are distinct and coprime to n, so they are all the residues coprime to n.

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Lemma 2.
If $\operatorname{ord}_n(a) = d$ then $\operatorname{ord}_n(a^j) = \frac{d}{\gcd(j,d)}$.

Lemma 3 (Gauss). $\sum_{d|m} \varphi(d) = m$ for every positive integer m.

Theorem 4. Let p be prime. For each $d \mid p-1$ there are exactly $\varphi(d)$ residues of order d modulo p. In particular there are $\varphi(p-1) \geq 1$ primitive roots.

Proof of Lemma 2. $(a^j)^t \equiv 1$ if and only if $d \mid jt$, if and only if $\frac{d}{\gcd(j,d)} \mid t$. □

Proof of Lemma 3. Write each of the fractions $\frac{1}{m}, \frac{2}{m}, \ldots, \frac{m}{m}$ in lowest terms as $\frac{a}{d}$. Then $d \mid m, 1 \leq a \leq d$ and $\gcd(a,d) = 1$, and every such fraction occurs exactly once. So the m fractions split into groups of sizes $\varphi(d)$, one for each $d \mid m$. □

Proof of Theorem 4. Let $\psi(d)$ be the number of residues of order d, for $d \mid p-1$. Suppose $\psi(d) \geq 1$ and let a have order d. Then $1, a, \ldots, a^{d-1}$ are d distinct roots of $x^d - 1$, which has at most d roots modulo p by Lagrange's theorem. So every element of order d is among them, and by Lemma 2 the element a^j has order d exactly when $\gcd(j,d) = 1$. Hence $\psi(d)$ is 0 or $\varphi(d)$. Every nonzero residue has some order dividing $p-1$, so

$$\sum_{d|p-1} \psi(d) = p - 1 = \sum_{d|p-1} \varphi(d),$$

using Lemma 3. Since $\psi(d) \leq \varphi(d)$ termwise, equality holds for every d. □

Note that the proof is a count, not a construction. No formula for a primitive root is known, and finding one in practice means testing candidates. By L4-NT1, g is a primitive root modulo p if and only if $g^{(p-1)/q} \not\equiv 1$ for every prime $q \mid p-1$.

Example 1.
Let $q > 3$ be a prime such that $p = 2q + 1$ is also prime. Prove that -3 is a primitive root modulo p.

Solution.
We cannot compute powers of -3 for general q, but $p - 1 = 2q$ has only two prime factors, so the test above needs only two checks, namely that a is a primitive root if and only if $a^2 \not\equiv 1$ and $a^q \not\equiv 1$. By Euler's criterion $a^q = a^{(p-1)/2} \equiv \left(\frac{a}{p}\right)$. So for $p \nmid a$ we get that *a is a primitive root if and only if $a \not\equiv \pm 1$ and a is a non-residue.*
Since $p \geq 11$, $-3 \not\equiv \pm 1$. From L5-NT1, $\left(\frac{-3}{p}\right) = \left(\frac{p}{3}\right)$. As $q > 3$ is prime, $q \equiv 1$ or $2 \pmod 3$. If $q \equiv 1$ then $3 \mid 2q + 1 = p$, impossible. So $q \equiv 2 \pmod 3$ and $p \equiv 5 \equiv 2 \pmod 3$, giving $\left(\frac{p}{3}\right) = -1$. Hence -3 is a non-residue and a primitive root. □
The same criterion shows that 3 is never a primitive root for such p, because $p \equiv 3 \pmod 4$ and so $\left(\frac{3}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{-3}{p}\right) = 1$. For $q = 5$ and $p = 11$ we have $-3 \equiv 8$, and $8^2 \equiv 9$, $8^5 \equiv 10 \equiv -1$, so the order of 8 is 10.

1.2 Using a generator

Fix a primitive root g modulo p. Every nonzero residue is g^i for an exponent i that is unique modulo $p-1$ (the *index*, or discrete logarithm), and multiplication of residues becomes addition of indices modulo $p-1$. Questions about powers become linear congruences.

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Theorem 5 (k-th powers).
Let $p \nmid a$ and $d = \gcd(k, p-1)$. Then $x^k \equiv a \pmod{p}$ has a solution if and only if $a^{(p-1)/d} \equiv 1$, and in that case the congruence has exactly d solutions. There are exactly $\frac{p-1}{d}$ nonzero k-th power residues.

Proof. Write $a = g^s$ and $x = g^t$. The congruence becomes $kt \equiv s \pmod{p-1}$, which is solvable if and only if $d \mid s$, and then has d solutions t modulo $p-1$. Also $a^{(p-1)/d} = g^{s(p-1)/d} \equiv 1$ if and only if $p-1 \mid s\frac{p-1}{d}$, if and only if $d \mid s$. The solvable s are the multiples of d, and there are $\frac{p-1}{d}$ of them modulo $p-1$. □

Some consequences. For $k = 2$ this is Euler's criterion. If $\gcd(k, p-1) = 1$ then $x \mapsto x^k$ is a bijection, so every integer is a cube modulo a prime $p \equiv 2 \pmod 3$. If $p \equiv 1 \pmod 3$ only a third of the nonzero residues are cubes. Since $g^{(p-1)/2}$ is a square root of 1 other than 1, we get $g^{(p-1)/2} \equiv -1$, and Wilson's theorem follows from $(p-1)! \equiv \prod_{j=0}^{p-2} g^j = \left(g^{(p-1)/2}\right)^{p-2} \equiv (-1)^{p-2} = -1$ for odd p.

1.3 Prime powers

Theorem 6.
Let p be an odd prime and g a primitive root modulo p.

1. If $g^{p-1} \not\equiv 1 \pmod{p^2}$, then g is a primitive root modulo p^k for every $k \geq 1$.
2. If $g^{p-1} \equiv 1 \pmod{p^2}$, then $g + p$ satisfies the condition in 1.
3. If h is a primitive root modulo p^k, then whichever of h, $h + p^k$ is odd is a primitive root modulo $2p^k$.

Proof. 1. Let $d = \operatorname{ord}_{p^k}(g)$. Then $g^d \equiv 1 \pmod p$, so $p-1 \mid d$, and $d \mid \varphi(p^k) = p^{k-1}(p-1)$. So $d = (p-1)p^j$ for some $j \leq k-1$. By LTE applied to g^{p-1} and 1 (valid since $p \mid g^{p-1} - 1$), $v_p\left(g^{(p-1)p^j} - 1\right) = v_p\left(g^{p-1} - 1\right) + j = 1 + j$. This is at least k only if $j = k-1$, so $d = \varphi(p^k)$.

2. $g + p$ is still a primitive root modulo p, and by the binomial theorem $(g+p)^{p-1} \equiv g^{p-1} + (p-1)pg^{p-2} \equiv 1 - pg^{p-2} \not\equiv 1 \pmod{p^2}$, since $p \nmid g$.

3. $\varphi(2p^k) = \varphi(p^k)$, and an odd residue has the same order modulo $2p^k$ as modulo p^k. □

Case 2 does happen, since 14 is a primitive root modulo 29 but $14^{28} \equiv 1 \pmod{29^2}$. For completeness, the full answer is that primitive roots modulo n exist if and only if n is 1, 2, 4, p^k or $2p^k$ with p an odd prime. The other direction is short. For $k \geq 3$ every odd x satisfies $x^{2^{k-2}} \equiv 1 \pmod{2^k}$ by the $p = 2$ form of LTE, and if $n = ab$ with $\gcd(a,b) = 1$ and $a, b > 2$, then $x^{\operatorname{lcm}(\varphi(a),\varphi(b))} \equiv 1 \pmod n$ with $\operatorname{lcm}(\varphi(a), \varphi(b)) < \varphi(n)$, because both totients are even.

1.4 Zsigmondy's theorem

Call a prime q a **primitive prime divisor** of $a^n - b^n$ if q divides $a^n - b^n$ but none of $a^k - b^k$ for $1 \leq k < n$.

Theorem 7 (Zsigmondy).

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Let $a > b \geq 1$ be coprime integers. Then $a^n - b^n$ has a primitive prime divisor for every $n \geq 1$, except in the cases (i) $n = 1$, $a - b = 1$, (ii) $n = 2$, $a + b$ a power of 2, (iii) $n = 6$, $a = 2$, $b = 1$. Similarly $a^n + b^n$ has a prime divisor dividing none of $a^k + b^k$ for $k < n$, except for $2^3 + 1^3 = 9$.

The theorem is due to Zsigmondy (1892), with the case $b = 1$ found earlier by Bang (1886). The proof factors $a^n - b^n$ into cyclotomic pieces and estimates their size, and takes a few pages. An elementary proof is in G. D. Birkhoff and H. S. Vandiver, "On the integral divisors of $a^n - b^n$", *Annals of Mathematics* 5 (1904), 173-180. I use the theorem here as a black box. A note on writing. Markers usually accept a correctly stated Zsigmondy, but a misquoted list of exceptions can cost most of the marks, and many problems are designed so that the case you need has a short proof by orders and LTE. When such a proof exists, I recommend writing that proof.

What a primitive prime divisor q gives you is that $q \nmid b$ (otherwise $q \mid a$ too), and the order of ab^{-1} modulo q is exactly n. Hence $n \mid q - 1$, so $q \equiv 1 \pmod{n}$ and $q > n$. A first consequence is that *for every n there are infinitely many primes $q \equiv 1 \pmod{n}$*. For each $m \geq 7$, take a primitive prime divisor q_m of $2^{nm} - 1$. Then $\operatorname{ord}_{q_m}(2) = nm$, so $q_m \equiv 1 \pmod{n}$, and different m give different orders, hence different primes.

Example 2.
Find all positive integers n such that every prime factor of $2^n - 1$ is less than 20.

Solution.
For small cases, $n = 1, 2, 3, 4$ give $1, 3, 7, 15$, all fine, and $n = 5$ gives 31, which fails. The obstruction is that $2^n - 1$ keeps acquiring new prime factors, and Zsigmondy says exactly that.
The orders of 2 modulo the primes below 20 are $3 \mapsto 2$, $5 \mapsto 4$, $7 \mapsto 3$, $11 \mapsto 10$, $13 \mapsto 12$, $17 \mapsto 8$, $19 \mapsto 18$. Suppose n works and let $d \mid n$ with $d \neq 1, 6$. Then $2^d - 1 \mid 2^n - 1$, and $2^d - 1$ has a primitive prime divisor q with $\operatorname{ord}_q(2) = d$. As $q < 20$, d is in the list above. So every divisor of n lies in $\{1, 2, 3, 4, 6, 8, 10, 12, 18\}$.
The only primes in this set are 2 and 3, and 9, 16, 24 are not in the set. So $n = 2^a 3^b$ with $a \leq 3$ and $b \leq 1$, and $(a, b) \neq (3, 1)$. This leaves $n \in \{1, 2, 3, 4, 6, 8, 12\}$. Conversely, $2^6 - 1 = 3^2 \cdot 7$, $2^8 - 1 = 3 \cdot 5 \cdot 17$ and $2^{12} - 1 = 3^2 \cdot 5 \cdot 7 \cdot 13$, so all seven values work.
Without Zsigmondy one can argue by size instead. By LTE, $v_q(2^n - 1) \leq v_q(2^e - 1) + v_q(n)$ where $e = \operatorname{ord}_q(2)$, so $2^n - 1$ is at most a fixed constant times a power of n, which fails for large n. That route needs explicit bounds on each v_q, which is more casework than quoting Zsigmondy.

1.5 Chinese remainder constructions

Many existence problems ("prove that there exist k consecutive integers such that ...") are solved by prescribing residues modulo many primes at once. The method in a nutshell:

1. Decide, for each required property, which congruence modulo which prime (or prime power) forces that property.
2. Use different primes for different conditions, so the moduli are pairwise coprime and the Chinese remainder theorem gives a solution, and adding multiples of the product gives infinitely many.
3. If the property needs primes of a special form (for instance $p \equiv 1 \pmod{4}$, so that -1 is a

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residue), first prove there are infinitely many of them.

Example 3.
Prove that there exist 2026 consecutive positive integers n such that, for each of them, $2^n - 1$ has at least 2026 distinct prime factors.

Solution.
First find a reason for $2^n - 1$ to have many prime factors. If $d \mid n$ then $2^d - 1 \mid 2^n - 1$, and $\gcd(2^a - 1, 2^b - 1) = 2^{\gcd(a,b)} - 1$. So if $q_1, \ldots, q_{2026}$ are distinct primes dividing n, the numbers $2^{q_i} - 1$ are pairwise coprime, each is greater than 1, and each divides $2^n - 1$. Choosing a prime factor of each gives 2026 distinct prime factors of $2^n - 1$.
So we only need to find N such that each of $N + 1, \ldots, N + 2026$ has at least 2026 distinct prime factors. Choose 2026^2 distinct primes and split them into 2026 groups of 2026, and let P_i be the product of the i-th group. The P_i are pairwise coprime, so by the Chinese remainder theorem there is a positive integer N with $N \equiv -i \pmod{P_i}$ for $i = 1, \ldots, 2026$. Then $P_i \mid N + i$, and we are done. □
Zsigmondy gives more. Each divisor $d \neq 1, 6$ of n contributes its own primitive prime divisor, so $2^n - 1$ has at least $\tau(n) - 2$ distinct prime factors. The gcd identity was enough here, and the identity is the safer thing to write.

1.6 Infinitely many primes p such that ...

The standard strategies:

1. (Euclid with a polynomial.) Let P be the product of the primes found so far and choose $N = f(P)$ so that every prime factor of N has the property and none divides P. For example every prime factor q of $(2P)^2 + 3$, where P is a product of primes $\equiv 1 \pmod 3$, satisfies $q \neq 2, 3$ and $\left(\frac{-3}{q}\right) = 1$, so $q \equiv 1 \pmod 3$.
2. (Residues of N.) If the property is a residue class, compute N modulo the modulus. If every prime factor were in the wrong class, N would be too.
3. (Growing orders.) Primitive prime divisors of $a^n - b^n$, or prime factors of $a^{2^k} + 1$, for growing n or k. Different orders give different primes.
4. Know what is out of reach. Dirichlet's theorem on primes in arithmetic progressions is deep, so quote the theorem only when no elementary route exists. "Infinitely many primes for which 2 is a primitive root" is Artin's conjecture and is still open.

Example 4.
Let k be a positive integer. Prove that there are infinitely many primes p such that each of $1, 2, \ldots, k$ is a quadratic residue modulo p.

Solution.
By multiplicativity of the Legendre symbol, we only need that 2 and every odd prime $q \leq k$ are residues. Each of these is a congruence condition on p, and all of them hold if $p \equiv 1$ modulo 8 and modulo every odd prime $q \leq k$.
Let $M = 8 \prod q$, the product over odd primes $q \leq k$, and let p be a prime with $p \equiv 1 \pmod M$ and $p > k$. By the consequence of Zsigmondy above there are infinitely many such p. Then $p \equiv 1 \pmod 8$, so $\left(\frac{2}{p}\right) = 1$. For an odd prime $q \leq k$, $p \equiv 1 \pmod 4$ and reciprocity give

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$\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right) = \left(\frac{1}{q}\right) = 1$. Every $m \le k$ is a product of such primes, so $\left(\frac{m}{p}\right) = 1$. □
For $k = 10$ the proof uses $p \equiv 1 \pmod{840}$, and the smallest such prime is 2521.

Common traps.

- To prove that g is a primitive root, check $g^{(p-1)/q} \not\equiv 1$ for every prime $q \mid p-1$, not just $g^{(p-1)/2}$.
- A primitive root modulo p need not be one modulo p^2 (the example of 14 and 29). Check the condition of Theorem 6.
- There is no primitive root modulo 8, since every odd square is 1 modulo 8.
- Zsigmondy needs $\gcd(a,b) = 1$, and the exceptions matter, since $2^6 - 1 = 3^2 \cdot 7$ has no primitive prime divisor, and neither has $3^2 - 1 = 8$.
- A primitive prime divisor is 1 modulo n, but is not necessarily the largest prime factor.
- The Chinese remainder theorem needs pairwise coprime moduli. Using the same prime for two conditions breaks the construction.

1.7 How to spot this in a contest

1. Sums or products over all residues ($\sum x^k$, $\prod x$, the set of k-th powers). Rewrite with a primitive root, and use Theorem 5 to count solutions of $x^k \equiv a$.
2. "Prove that g is a primitive root" or "find the order exactly". Test the primes dividing $p-1$, and when $p - 1 = 2q$ this reduces to a Legendre symbol.
3. Equations in $a^n - b^n$ or $a^n + 1$ where all prime factors must come from a small fixed set. Zsigmondy bounds n, or orders plus LTE do this by hand.
4. "There exist k consecutive integers", "there exists n with many prime factors", "infinitely many n". Use a Chinese remainder construction.
5. "Infinitely many primes with ...". Use Euclid with a well chosen polynomial, or growing orders.

2 Practice problems

The problems are roughly in order of difficulty, from ⋆⋆ to ⋆⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

Problem 1 ⋆⋆ Find all primes p such that p divides $1^{2026} + 2^{2026} + \cdots + (p-1)^{2026}$. (You may use that 1013 and 2027 are prime.)

Problem 2 ⋆⋆ Let p be an odd prime and g a primitive root modulo p. Prove that $-g$ is a primitive root modulo p if and only if $p \equiv 1 \pmod 4$.

Problem 3 ⋆⋆ Prove that there are infinitely many primes p with $p \equiv 3 \pmod 8$.

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Problem 4 ⋆⋆⋆ Let p be an odd prime and a an integer not divisible by p. Prove that the congruence $x^p \equiv a \pmod{p^2}$ has a solution if and only if $a^{p-1} \equiv 1 \pmod{p^2}$.

Problem 5 ⋆⋆⋆ Find all positive integers n such that $2^n + 1$ has no prime factor greater than 7. Give a proof that does not use Zsigmondy's theorem.

Problem 6 ⋆⋆⋆⋆ Prove that there are infinitely many primes q such that q does not divide $2^n + 3^n$ for any positive integer n.

3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.

1. **IMO 1971 Problem 3** ⋆⋆
   Build the subset one term at a time. Euler's theorem tells you which exponent to choose next.
2. **IMO 1989 Problem 5** ⋆⋆
   Chinese remainder theorem with two primes per number.
3. **UKMT BMO2 2003/04 Problem 2** ⋆⋆
   Powers of 2 repeat modulo the odd part of 2004, so choose the exponents of the 1s in a convenient residue class.
4. **USAMO 2017 Problem 1** ⋆⋆
   Look for a one-parameter family where $a + b$ is small compared with a and b and the powers are easy to reduce.
5. **UKMT BMO2 2007/08 Problem 4** ⋆⋆⋆
   Try a family with y a fixed multiple or polynomial in x and solve the divisibility.
6. **UKMT BMO2 2012/13 Problem 1** ⋆⋆⋆
   From one solution, produce a larger one by replacing a variable with the other root of a quadratic.
7. **USAMO 2005 Problem 2** ⋆⋆⋆
   Find a prime modulo which the high powers in the system take very few values. Theorem 5 tells you which prime.
8. **IMO Shortlist 2002 N3** ⋆⋆⋆
   Find many pairwise coprime factors of $2^m + 1$ using the divisors of m, or use Zsigmondy for each divisor.
9. **IMO 2000 Problem 5** ⋆⋆⋆⋆
   Decide the answer by experimenting with powers of 3, then add primes one at a time by induction.
10. **IMO 2003 Problem 6** ⋆⋆⋆⋆
    Choose q so that p-th powers modulo q are restricted. Theorem 5 says which q make this happen.

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4 Hints

1. By Theorem 1, the sum is 0 or -1 modulo p. Which p have $p-1 \mid 2026$?
2. Write -1 as a power of g and use Lemma 2.
3. Consider P^2+2, where P is the product of finitely many such primes. Which primes can divide P^2+2, and what is P^2+2 modulo 8?
4. One direction is Euler's theorem. For the other, use a primitive root modulo p^2 and write the congruence in terms of indices.
5. Show 7 never divides 2^n+1, and that 3 and 5 cannot both divide 2^n+1. Then use LTE to show 2^n+1 would be too small.
6. $q \mid 2^n+3^n$ means $(2\cdot 3^{-1})^n \equiv -1$. This is impossible if the order of $2\cdot 3^{-1}$ is odd. When is that guaranteed for $q \equiv 3 \pmod 4$?

5 Solutions

Problem 1

Find all primes p such that p divides $1^{2026}+2^{2026}+\cdots+(p-1)^{2026}$. (You may use that 1013 and 2027 are prime.)

By Theorem 1 the sum is $\equiv -1 \pmod p$ if $p-1 \mid 2026$ and $\equiv 0$ otherwise. (For $p=2$ the sum is $1 \equiv -1$, consistent with $1 \mid 2026$.) Since $p \geq 2$, $-1 \not\equiv 0$, so p divides the sum if and only if $p-1 \nmid 2026$.

Now $2026 = 2\cdot 1013$ with 1013 prime, so its divisors are $1, 2, 1013, 2026$, and $p-1 \mid 2026$ means $p \in \{2, 3, 1014, 2027\}$. Of these 1014 is even and not prime, while 2, 3, 2027 are prime. So p divides the sum for every prime except 2, 3 and 2027.

Answer: All primes except 2, 3 and 2027.

Problem 2

Let p be an odd prime and g a primitive root modulo p. Prove that $-g$ is a primitive root modulo p if and only if $p \equiv 1 \pmod 4$.

Since g has order $p-1$, the residue $t = g^{(p-1)/2}$ satisfies $t^2 \equiv 1$ and $t \not\equiv 1$. The only square roots of 1 modulo a prime are ± 1, so $t \equiv -1$. Hence $-g \equiv g^{(p+1)/2}$, and by Lemma 2 its order is $\frac{p-1}{\gcd\left(\frac{p+1}{2}, p-1\right)}$. So $-g$ is a primitive root if and only if $e = \gcd\left(\frac{p+1}{2}, p-1\right) = 1$.

Any common divisor of $\frac{p+1}{2}$ and $p-1$ divides $p+1$ and $p-1$, hence divides 2. So $e \in \{1, 2\}$, and since $p-1$ is even, $e=2$ if and only if $\frac{p+1}{2}$ is even, that is, $p \equiv 3 \pmod 4$. Therefore $e=1$ if and only if $p \equiv 1 \pmod 4$, which proves both directions. $\square$

Problem 3

Prove that there are infinitely many primes p with $p \equiv 3 \pmod 8$.

Suppose there are only finitely many such primes, and let P be their product ($P=1$ if there are none). Then P is odd, so $P^2 \equiv 1 \pmod 8$ and $N = P^2+2 \equiv 3 \pmod 8$. In particular N is odd

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and $N \geq 3$.

Let q be a prime factor of N. Then q is odd, and $q \nmid P$, since otherwise $q \mid 2$. From $P^2 \equiv -2 \pmod{q}$ with $q \nmid P$, -2 is a quadratic residue modulo q, so by L5-NT1 (Theorem 4 there) $q \equiv 1$ or $3 \pmod 8$. If every prime factor of N were 1 modulo 8, then N would be 1 modulo 8, which is false. So N has a prime factor $q \equiv 3 \pmod 8$, and $q \nmid P$, so q is not on our list. This contradiction proves the claim. □

Problem 4

Let p be an odd prime and a an integer not divisible by p. Prove that the congruence $x^p \equiv a \pmod{p^2}$ has a solution if and only if $a^{p-1} \equiv 1 \pmod{p^2}$.

Suppose $x^p \equiv a \pmod{p^2}$. Then $p \nmid x$, as $p \nmid a$. By Euler's theorem, since $\varphi(p^2) = p(p-1)$, we get $a^{p-1} \equiv x^{p(p-1)} \equiv 1 \pmod{p^2}$.

Conversely, suppose $a^{p-1} \equiv 1 \pmod{p^2}$. By Theorem 4 and Theorem 6 there is a primitive root g modulo p^2, of order $p(p-1)$. Write $a \equiv g^s \pmod{p^2}$. Then $g^{s(p-1)} \equiv 1$, so $p(p-1) \mid s(p-1)$, that is, $p \mid s$. Write $s = pt$ and let $x = g^t$. Then $x^p = g^{pt} = g^s \equiv a \pmod{p^2}$, so a solution exists. □

Remark. The same index argument shows that when a solution exists there are exactly p of them modulo p^2, since $pt \equiv s \pmod{p(p-1)}$ has $\gcd(p, p(p-1)) = p$ solutions t.

Problem 5

Find all positive integers n such that $2^n + 1$ has no prime factor greater than 7. Give a proof that does not use Zsigmondy's theorem.

The powers of 2 modulo 7 are $2, 4, 1$ repeating, never 6, so $7 \nmid 2^n + 1$. Since $2^n + 1$ is odd, the condition says $2^n + 1 = 3^a 5^b$ for some $a, b \geq 0$. Also $3 \mid 2^n + 1$ if and only if n is odd (as $2 \equiv -1 \pmod 3$), and $5 \mid 2^n + 1$ if and only if $n \equiv 2 \pmod 4$ (as $\operatorname{ord}_5(2) = 4$ and $2^2 \equiv -1$).

Case n odd. Then $5 \nmid 2^n + 1$, so $2^n + 1 = 3^a$. Since n is odd, $2^n + 1 = 2^n - (-1)^n$ with $3 \mid 2 - (-1)$, and LTE gives $a = v_3(3) + v_3(n) = 1 + v_3(n)$. Hence $2^n + 1 = 3 \cdot 3^{v_3(n)} \leq 3n$. But $2^n + 1 > 3n$ for $n \geq 4$ (true at $n = 4$, and the left side more than doubles while the right side grows by 3). So $n \in \{1, 3\}$, and both work, giving 3 and 9.

Case n even. Then $3 \nmid 2^n + 1$, so $2^n + 1 = 5^b$ with $b \geq 1$, which forces $n \equiv 2 \pmod 4$. Write $n = 2m$ with m odd. Then $2^n + 1 = 4^m - (-1)^m$ with $5 \mid 4 - (-1)$, and LTE gives $b = 1 + v_5(m)$, so $4^m + 1 = 5 \cdot 5^{v_5(m)} \leq 5m$. But $4^m + 1 > 5m$ for $m \geq 2$, so $m = 1$, $n = 2$, and $2^2 + 1 = 5$ works.

Hence $n \in \{1, 2, 3\}$. (With Zsigmondy, $2^n + 1$ has a prime factor not dividing $2^k + 1$ for any $k < n$ unless $n = 3$. Since $3 \mid 2^1 + 1$ and $5 \mid 2^2 + 1$, for $n \geq 4$ that prime is at least 11.)

Answer: $n = 1, 2, 3$

Problem 6

Prove that there are infinitely many primes q such that q does not divide $2^n + 3^n$ for any positive integer n.

Step 1 (a sufficient condition). Let $q > 3$ be a prime with $q \equiv 3 \pmod 4$ and $\left(\frac{6}{q}\right) = 1$. Let $t \equiv 2 \cdot 3^{-1} \pmod q$. Since $t \equiv 6 \cdot (3^{-1})^2$, multiplicativity gives $\left(\frac{t}{q}\right) = \left(\frac{6}{q}\right) = 1$, so by Euler's criterion $t^{(q-1)/2} \equiv 1$ and $\operatorname{ord}_q(t)$ divides $\frac{q-1}{2}$, which is odd because $q \equiv 3 \pmod 4$. If $q \mid 2^n + 3^n$ for some n, then multiplying by $(3^{-1})^n$ gives $t^n \equiv -1$, so $t^{2n} \equiv 1$ while $t^n \not\equiv 1$. Then $\operatorname{ord}_q(t)$ divides $2n$ but

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not n, so this order is even. This is a contradiction. Hence such q never divides $2^n + 3^n$.

Step 2 (infinitely many such primes). Suppose there are only finitely many primes as in Step 1, and let P be their product ($P = 1$ if there are none). Let $N = 24P^2 - 1$. Then $N \geq 23$, $N \equiv -1 \pmod 4$, and $N \equiv -1 \pmod 2$ and $\pmod 3$, so no prime factor of N is 2 or 3. Since N is positive and $N \equiv 3 \pmod 4$, N has a prime factor $q \equiv 3 \pmod 4$. Then $q > 3$, and $q \nmid P$ since otherwise $q \mid 1$. From $24P^2 \equiv 1 \pmod q$ we get $6 \cdot (2P)^2 \equiv 1$, so $\left(\frac{6}{q}\right) = \left(\frac{6(2P)^2}{q}\right) = \left(\frac{1}{q}\right) = 1$. So q satisfies the conditions of Step 1 but does not divide P, a contradiction.

Combining the two steps, there are infinitely many primes q that divide no number $2^n + 3^n$. $\square$

Remark. Starting from $P = 1$, the construction finds 23, then (with $P = 23$) the prime 2539, and so on. One checks directly that the order of $2 \cdot 3^{-1} \equiv 16$ modulo 23 is 11.